

# DATA VISUALIZATION & BUSINESS INTELLIGENCE

## MASTER EXAM PREP GUIDE

■ Complete Beginner-to-Expert Coverage | ■ Exam-Focused | ■ Project Deep-Dive

|                    |                                                      |
|--------------------|------------------------------------------------------|
| ■ Source Material  | Research Papers, ICA Report, Slides, Project PDFs    |
| ■ Exam Pattern     | 8-10 Questions: Theory + Comparisons + Project-based |
| ■ Difficulty Level | Beginner-friendly → Exam-ready                       |
| ■ Special Focus    | XYZ Bank Customer Churn Project (Power BI)           |

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# PART 1: BUSINESS INTELLIGENCE – FOUNDATIONS

## 1.1 What is Business Intelligence (BI)?

■ **Simple Analogy:** *Imagine you run a shop. Every day you write down what you sold, to whom, and when. After a month, you look at those papers to figure out what sells most, who your best customers are, and when to order more stock. THAT is Business Intelligence — using past data to make smarter decisions!*

**Technical Definition:** Business Intelligence (BI) is a set of strategies, technologies, and tools that collect, integrate, analyze, and present business data to help organizations make better decisions.

BI answers questions like: *What happened? Why did it happen? What is likely to happen next?*

### ■ Key Components of BI:

| Component      | Simple Meaning                                | Example                                          |
|----------------|-----------------------------------------------|--------------------------------------------------|
| Data Source    | Where data comes from                         | Bank transactions, Excel files, databases        |
| ETL Process    | Extract, Transform, Load — clean & move data  | Getting CSV from Kaggle, cleaning in Power Query |
| Data Warehouse | A big storage for organized business data     | SQL Server database storing all bank records     |
| OLAP           | Online Analytical Processing — fast analysis  | Slicing bank data by country, age, tenure        |
| Dashboard      | Visual display of key numbers                 | XYZ Bank Power BI dashboard                      |
| KPI            | Key Performance Indicator — a success measure | Churn Rate = 20.37%                              |
| Reports        | Detailed outputs for decision makers          | Customer Churn Impact Report                     |

■ **IMPORTANT FOR EXAM:** BI is NOT the same as Data Science. BI looks at past/present data to report insights. Data Science predicts future data using machine learning.

## 1.2 Traditional BI vs Modern BI (Agile BI)

■ **MOST EXPECTED QUESTION:** What is Agile BI and why is it needed?

Traditional BI was slow — it needed IT professionals to build OLAP cubes, and any new question required rebuilding everything. Modern (Agile) BI allows fast, flexible, self-service analytics.

| Feature      | Traditional BI            | Agile / Modern BI                        |
|--------------|---------------------------|------------------------------------------|
| Speed        | Days/weeks to get answers | Minutes/real-time                        |
| Who uses it? | Only IT / specialists     | Business users themselves (self-service) |
| Data sources | One relational database   | Databases, files, cloud, OData, Excel    |
| Flexibility  | Rigid — pre-built cubes   | Flexible — ad-hoc queries                |
| Technology   | OLAP cubes (MDX)          | In-memory columnar (DAX, xVelocity)      |

| Feature      | Traditional BI        | Agile / Modern BI                   |
|--------------|-----------------------|-------------------------------------|
| Example tool | SSAS Multidimensional | Power BI Tabular Model, Power Pivot |

■ **REMEMBER THIS:** The research paper says 3 criteria for Agile BI: (1) Integrate new/unforeseen data sources, (2) Rapidly embrace change, (3) Provide self-service BI.

## 1.3 OLAP – What is it?

■ **Analogy:** Think of a Rubik's cube. Each face is a different 'dimension' of your data (time, product, location). OLAP lets you rotate that cube to see any combination of data quickly.

**OLAP** = Online Analytical Processing. It allows users to analyze large volumes of data from multiple perspectives (dimensions) at high speed.

| OLAP Type | Full Name             | Storage                        | Speed        | Use Case                       |
|-----------|-----------------------|--------------------------------|--------------|--------------------------------|
| MOLAP     | Multidimensional OLAP | Pre-aggregated cubes in memory | Fastest      | Fixed, repeated queries        |
| ROLAP     | Relational OLAP       | Queries live relational DB     | Slower       | Large data, flexible queries   |
| HOLAP     | Hybrid OLAP           | Mix of both                    | Medium       | Balance of speed & flexibility |
| DOLAP     | Desktop OLAP          | Local machine                  | Fast locally | Personal analysis              |

■ **COMMON MISTAKE:** Students confuse OLAP with OLTP. OLTP (Online Transaction Processing) is for daily transactions (like ATM withdrawals). OLAP is for analysis (like 'how many withdrawals happened in France last month?').

## 1.4 The BI Semantic Model (BISM) — From Research Paper

Microsoft's **BISM** combines two data models under one roof:

- **Multidimensional Model:** Old approach using MDX language and ROLAP/MOLAP storage
- **Tabular Model:** New approach using DAX language and xVelocity/Direct Query storage

BISM integrates data from: Databases, LOB Apps, Files, OData Feeds, Cloud Services — and delivers to: BI Apps, Reports, Excel, PowerPivot, Dashboards.

■ **PRO TIP:** The Tabular Model was introduced in SQL Server 2012. It uses xVelocity (in-memory columnar storage) for ultra-fast queries without pre-built indexes.

## PART 2: POWER BI vs TABLEAU – COMPLETE COMPARISON

### 2.1 What is Power BI?

■ **Analogy:** Power BI is like a super-smart Excel that can connect to any data, clean it, show beautiful charts, and automatically refresh every day.

**Power BI** is a business analytics service by Microsoft. It provides interactive visualizations with an interface simple enough for end-users to create reports and dashboards. It allows users to connect to multiple data sources, clean data (Power Query), create calculations (DAX), and share insights.

**Access Options:** Web browser, mobile device, desktop application. Integrates with Excel, SharePoint, Teams, Azure.

### 2.2 What is Tableau?

**Tableau** is a data visualization tool that enables users to connect, visualize and share data with a drag-and-drop interface. No coding or programming skills required. Strong on charts: line charts, bar charts, maps, scatter plots, etc.

**Tableau Suite:** Tableau Desktop (create visuals), Tableau Server (share/collaborate), Tableau Prep (clean data).

### 2.3 Power BI vs Tableau – Detailed Comparison

■ **MOST EXPECTED QUESTION: Compare Power BI and Tableau.**

| Feature          | Tableau                                                | Power BI                                             |
|------------------|--------------------------------------------------------|------------------------------------------------------|
| Cost             | High cost, enterprise-level                            | Free version + affordable Pro license                |
| Ease of Use      | Intuitive drag-and-drop, steeper learning for advanced | User-friendly, easy Microsoft ecosystem integration  |
| Data Integration | Wide range: on-premises and cloud                      | Excellent Microsoft + variety of other sources       |
| Performance      | Strong with large datasets, resource-intensive         | Good, optimized for Microsoft Azure                  |
| Visualization    | Extensive, highly customizable                         | Robust, slightly less flexible than Tableau          |
| AI Features      | Tableau AI, R and Python integration                   | Power BI Q&A, AI visuals, Azure ML integration       |
| Deployment       | On-premises, cloud, hybrid                             | Cloud-based + on-premises via Power BI Report Server |
| Mobile Support   | Strong, native apps                                    | Excellent mobile optimization and native apps        |
| Collaboration    | Tableau Server and Online                              | Microsoft Teams integration                          |
| Data Prep        | Tableau Prep (separate product)                        | Built-in Power Query (integrated)                    |

| Feature            | Tableau                                         | Power BI                                        |
|--------------------|-------------------------------------------------|-------------------------------------------------|
| Community          | Large active community                          | Strong Microsoft support community              |
| Scripting Language | Not applicable / Python/R                       | DAX + M-Language (Power Query)                  |
| Best For           | Sophisticated visualizations, large enterprises | Microsoft ecosystem users, cost-conscious teams |

■ **MEMORY TRICK: Power BI = Price-friendly, Integrates with Microsoft. Tableau = Top visuals, Amazing customization, Big price tag.**

■ **IMPORTANT FOR EXAM: Both tools are BI platforms. Key differences: Cost (Power BI wins), Customization (Tableau wins), Microsoft integration (Power BI wins), Learning curve (Power BI easier).**

## PART 3: IMPORT vs DIRECT QUERY vs LIVE CONNECTION

■ **MOST EXPECTED QUESTION:** What is the difference between Import, Direct Query, and Live Connection in Power BI?

### 3.1 Import Mode

■ **Analogy:** You take a photo of a whiteboard. You have a copy of everything — fast to look at later — but if the whiteboard changes, your photo is outdated.

**Import** is the DEFAULT mode. Data is copied into Power BI's in-memory storage (compressed). You work with your own local copy.

**When to use Import:** Smaller to medium datasets that fit in memory. When you need extensive transformations, calculated columns, and measures. When you want the fastest query response.

**Advantages:** Fastest performance. Full DAX capabilities. Works offline. Full Power Query transformation support.

**Disadvantages:** Data is NOT live — needs scheduled refresh. Large datasets can be slow to load.

### 3.2 Direct Query Mode

■ **Analogy:** Instead of taking a photo, you look at the whiteboard through a live video call. Every time you look, you see exactly what's on the board right now.

**Direct Query** does NOT import data. Every chart/visual sends a live query to the database. Data is always current.

**When to use Direct Query:** Very large datasets that can't fit in memory. When real-time or near-real-time data is needed. Transactional databases, streaming data.

**Advantages:** Always live/current data. No data size limit.

**Disadvantages:** Slower performance (every interaction = a new DB query). Limited DAX functions. Requires database to be available.

### 3.3 Live Connection Mode

■ **Analogy:** You are using your company's already-built library catalog system. You don't copy the books — you just read them directly from the central library.

**Live Connection** connects to an existing Analysis Services model (SSAS or Power BI Premium dataset). Power BI acts only as a visualization layer — no data import, no transformation.

**When to use Live Connection:** Organizations with existing SSAS models. Multiple teams sharing one central data model. Consistent definitions across all reports.

### 3.4 Comparison Table

| Feature             | Import               | Direct Query | Live Connection    |
|---------------------|----------------------|--------------|--------------------|
| Data stored in PBI? | YES (copy)           | NO           | NO                 |
| Data freshness      | Scheduled refresh    | Real-time    | Real-time          |
| Performance         | Fastest              | Slower       | Good (server-side) |
| Dataset size limit  | ~1 GB (Power BI Pro) | No limit     | No limit           |

| Feature                 | Import                 | Direct Query         | Live Connection          |
|-------------------------|------------------------|----------------------|--------------------------|
| Full DAX support?       | YES (full)             | LIMITED              | LIMITED                  |
| Power Query transforms? | Full support           | Limited              | Not supported            |
| Offline use?            | YES                    | NO                   | NO                       |
| Best for                | Small-medium datasets  | Large/real-time data | Shared enterprise models |
| Connection to           | Any source then copies | Single DB/source     | SSAS / PBI Premium       |

■ **COMMON MISTAKE:** Students confuse Direct Query with Live Connection. Key difference: Direct Query can connect to many source types and you create the model in Power BI. Live Connection uses a model that **ALREADY EXISTS** in Analysis Services.

■ **MEMORY TRICK:** Import = I copy it. Direct = Direct to source every time. Live = Link to existing model.



## PART 4: VIEWS vs TABLES IN POWER BI

### 4.1 What is a Table?

A **table** is what you normally think of — rows and columns of data stored physically in a database. It holds actual data on disk.

### 4.2 What is a View?

■ **Analogy:** A Table is like an actual filing cabinet full of papers. A View is like a window that lets you SEE specific papers from that cabinet — but it doesn't copy them. Every time you look through the window, you see the current state of the cabinet.

A **database view** is a stored query (saved SQL SELECT statement) over table data. It does NOT physically store data — every time you access it, it runs the query and shows current results.

### 4.3 Three Reasons to Use Views Instead of Tables (from your notes)

■ **MOST EXPECTED QUESTION:** Give 3 reasons why views are preferred over tables in Power BI.

**Reason 1: Simplify Complexity:** Views can join multiple tables into one result set. Instead of importing 5 tables and joining in Power Query, you create one view that already has everything.

**Reason 2: Data Security & Access Control:** You can create a view showing only US sales data and give users access to the view only — not the underlying table with ALL global data.

**Reason 3: Ensure Consistency:** All reports see the same column names and data. If you rename 'BrandName' to 'Brand' in the view, EVERY report using that view gets 'Brand' automatically — no manual renaming in each report.

### 4.4 Other Advantages of Views

- Can filter data (e.g., only sales > £1000)
- Can pre-aggregate data using SUM, AVG, COUNT
- Don't consume physical space in the database
- Always return current data (no stale snapshots)
- User-friendly column naming (no 'DimProduct', 'ETLLoadID')

### Views vs Tables Comparison:

| Feature               | Table                     | View                                        |
|-----------------------|---------------------------|---------------------------------------------|
| Physical data stored? | YES                       | NO — runs query each time                   |
| Data freshness        | Static until updated      | Always current                              |
| Join multiple tables? | NO (single table)         | YES — can join many tables                  |
| Access control?       | Whole table or nothing    | Fine-grained — show only what's needed      |
| Space usage           | Takes disk space          | Only stores query definition (trivial)      |
| Performance           | Faster (direct access)    | Slightly slower (re-executes query)         |
| Best for              | Simple, direct data loads | Complex joins, security, naming consistency |

■ **SHORT NOTE:** Materialized/Indexed Views are the best of both worlds — they store the result physically (like a table) but have view logic (joins, transforms). Trade-off: they consume storage but give performance benefits.

## PART 5: NATIVE DATABASE QUERY IN POWER BI

### 5.1 What is a Native Database Query?

■ **Analogy:** *Instead of using Power BI's visual point-and-click to pick tables, you write a direct SQL letter to the database saying exactly what you want. It bypasses the normal import process.*

A **Native Database Query** allows you to write raw SQL (or database-specific language) directly in Power BI when connecting to a database. It lets you get precisely the data you need in one step.

### 5.2 Steps to Use Native Database Query:

**Step 1:** Go to: Data > Get Data > From Databases > From SQL Server Database

**Step 2:** Specify the Server name and Database name

**Step 3:** Click Advanced Options

**Step 4:** Type your SQL query in the SQL Statement box

**Step 5:** Click OK

**Step 6:** First connection: choose authentication mode (Windows/Database)

**Step 7:** Data loads into Power Query Editor

**Step 8:** Shape the data as needed, then click Apply & Close

### 5.3 Real Example from Your Notes (SQL CASE Statement):

The SQL query from your notes used a CASE WHEN statement to transform the 'Status' column into business-friendly categories like 'Resulted', 'NewResults', 'PendingOrder', 'NewOrder', 'SentToBilling', 'TotalMoneyCollected'. This is a classic real-world use of native queries.

```
SELECT CASE WHEN m.[Status] IN ('validated') THEN 'Resulted' WHEN m.[Status] IN ('results available') THEN 'NewResults' WHEN m.[Status] IN ('processing') THEN 'PendingOrder' WHEN m.[Status] IN ('Ready to ship') THEN 'NewOrder' WHEN m.[Status] IN ('Sent to Billing') THEN 'SentToBilling' ELSE m.[Status] END AS Status, COUNT(m.[Status]) AS StatusCount FROM MasterRequisitionsInfo AS m WHERE m.STATUS NOT IN ('deleted')
```

■ **IMPORTANT SECURITY WARNING: Running a Native Database Query from ANOTHER USER may alter your database. Power BI shows a warning prompt: 'Do you approve running this native query? Native queries may be unsafe and alter the database.'**

■ **EXAM TIP:** The benefit of native queries is saving time — you get exactly the data you need in one query instead of importing everything and then filtering/transforming in Power Query.

## PART 6: DATA MODELING – STAR & SNOWFLAKE SCHEMAS

### 6.1 What is a Data Model?

■ **Analogy:** A data model is like the blueprint of a house. It shows which rooms (tables) exist, how they connect (doors = relationships), and what's in each room (columns). Without a good blueprint, the house collapses!

A **data model** organizes tables and defines relationships between them. In Power BI, the Model View shows all tables and their connections. Good modeling = fast, accurate reports.

### 6.2 Fact Table vs Dimension Table

■ **MOST EXPECTED QUESTION:** What is the difference between a Fact Table and a Dimension Table?

■ **Analogy:** In a bank, the **FACT TABLE** is like the transaction log — every individual transaction. **DIMENSION TABLES** are like lookup books — customer information, country details, product types.

| Property    | Fact Table                                     | Dimension Table                                          |
|-------------|------------------------------------------------|----------------------------------------------------------|
| Contains    | Numerical measurements/events                  | Descriptive attributes                                   |
| Examples    | Balance, CreditScore, Salary, Sales Amount     | Customer, Product, Geography, Time, Gender               |
| Size        | Usually very LARGE (millions of rows)          | Usually small (hundreds/thousands of rows)               |
| Keys        | Foreign Keys (linking to dimensions)           | Primary Key                                              |
| Data type   | Mostly numbers / measures                      | Mostly text / categories                                 |
| In project  | Churn_Modelling (Balance, CreditScore, Salary) | HasCredit, IsActive, Churn, Gender, Geography, Age_Group |
| Aggregation | YES — sum, average, count                      | NO — used for grouping/filtering                         |

### 6.3 Star Schema vs Snowflake Schema

■ **Star Analogy:** Draw a star — the center is the Fact Table, each point is a Dimension Table. All dimensions connect **DIRECTLY** to the fact table. Simple and fast.

■ **Snowflake Analogy:** Like a star, but each 'point' has **MORE** points branching off it. Dimensions connect to other dimensions. More complex but saves storage.

| Feature          | Star Schema                        | Snowflake Schema                         |
|------------------|------------------------------------|------------------------------------------|
| Structure        | Fact table + flat dimension tables | Fact table + normalized dimension tables |
| Dimension tables | NOT normalized (flat)              | Normalized (split into sub-tables)       |
| Query speed      | Faster (fewer joins)               | Slower (more joins needed)               |
| Storage          | More storage (data duplication)    | Less storage (normalization)             |
| Complexity       | Simpler to understand              | More complex                             |

| Feature          | Star Schema                          | Snowflake Schema                                 |
|------------------|--------------------------------------|--------------------------------------------------|
| Used in project? | Base structure                       | YES — Age_Group is sub-dimension of Age & Tenure |
| Example          | Churn_Modelling + HasCredit + Gender | Fact → Age&Tenure → Age_Group (bridged)          |

■ **REMEMBER:** In the XYZ Bank project, the schema is SNOWFLAKE because Age\_Group is attached to Age & Tenure table, which is then attached to the Fact Table. This creates a bridge/chain connection — that's snowflake!

## 6.4 Relationships in Power BI

**One-to-Many (1:\*)** — Most common. One row in Dimension matches MANY rows in Fact.

Example: One 'Gender' row (Female) links to many customers who are female in Churn\_Modelling.

**One-to-One (1:1)** — One row matches exactly one other row.

Example: Age & Tenure table has 1:1 with Fact Table via CustomerID.

■ **COMMON MISTAKE:** Never create Many-to-Many relationships directly in Power BI — they cause ambiguity. Use a bridge table instead.

■ **EXAM TIP:** Relationships work through KEY columns. Primary Key in Dimension, Foreign Key in Fact. Think of it like an electrical connection — if the plugs don't match, no power flows!

## PART 7: DAX – DATA ANALYSIS EXPRESSIONS (COMPLETE GUIDE)

### 7.1 What is DAX?

■ **Analogy:** DAX is like Excel formulas, but 1000x more powerful. Instead of working on one cell, DAX works on entire columns and tables in Power BI's data model.

**DAX** (Data Analysis Expressions) is a formula language used in Power BI, Power Pivot, and SQL Server Analysis Services to define calculated columns and measures. It was created alongside the Tabular Model and is optimized for multi-core processors.

**Two main uses:** (1) Measures — calculated at query time (dynamic), (2) Calculated Columns — added to a table permanently.

### 7.2 Measures vs Calculated Columns

■ **MOST EXPECTED QUESTION:** Difference between a Measure and a Calculated Column in DAX.

| Property            | Measure                           | Calculated Column                            |
|---------------------|-----------------------------------|----------------------------------------------|
| When calculated?    | At query time (dynamic)           | At data refresh time (static)                |
| Stored in table?    | NO — lives in model               | YES — added as new column to table           |
| Row context?        | NO (uses filter context)          | YES (calculates row by row)                  |
| Affects model size? | NO                                | YES (adds memory)                            |
| Example             | Churn Rate = [Exited]/[Customers] | Salary_Group = Low/Mid/High based on salary  |
| Aggregation         | YES — SUM, AVERAGE, COUNT         | NO aggregation                               |
| Usage               | KPIs, totals, percentages         | Grouping, categorizing, text transformations |

### 7.3 ALL DAX Functions from the XYZ Bank Project

| # | Function         | Syntax                                      | Purpose                                   | Example from Project                                         |
|---|------------------|---------------------------------------------|-------------------------------------------|--------------------------------------------------------------|
| 1 | DISTINCTCOUNT    | DISTINCTCOUNT(Table[Column])                | Counts unique values only                 | Customers = DISTINCTCOUNT(Churn_Modelling[CustomerId])       |
| 2 | CALCULATE        | CALCULATE(Expression, Filter1, Filter2,...) | Evaluates expression with applied filters | Exited = CALCULATE([Customers], Churn[ChurnStatus]='Exited') |
| 3 | SUM              | SUM(Table[Column])                          | Adds all values in a column               | Total Balance = SUM(Churn_Modelling[Balance])                |
| 4 | AVERAGE          | AVERAGE(Table[Column])                      | Calculates arithmetic mean                | Avg. CreditScore = AVERAGE(Churn_Modelling[CreditScore])     |
| 5 | DIVIDE (implied) | [Part]/[Whole]                              | Calculates percentage                     | Churn Rate = [Exited]/[Customers]                            |

| # | Function | Syntax                        | Purpose                                                           | Example from Project                                   |
|---|----------|-------------------------------|-------------------------------------------------------------------|--------------------------------------------------------|
| 6 | VALUES   | VALUES(Table[Column])         | Returns distinct values from a column — used for reference tables | Salary_Group uses VALUES('Salary Range'[Salary_Range]) |
| 7 | VAR      | VAR VariableName = expression | Stores a value in a variable for reuse                            | VAR Class = 'Age & Tenure'[Customer_Tenure]            |
| 8 | RETURN   | RETURN Expression             | Returns result of a VAR block                                     | RETURN<br>CALCULATE(VALUES(...), conditions)           |

## 7.4 All Measures Created in the Project

| S/N | Measure Name          | DAX Formula                                            | What it does                                     |
|-----|-----------------------|--------------------------------------------------------|--------------------------------------------------|
| 1   | Customers             | DISTINCTCOUNT(Churn_Modelling[CustomerId])             | Total unique customers (10,000)                  |
| 2   | Exited                | CALCULATE([Customers], Churn[ChurnStatus]='Exited')    | Count of customers who left (2,037)              |
| 3   | Retained              | CALCULATE([Customers], Churn[ChurnStatus]='Retained')  | Count of customers still with bank (7,963)       |
| 4   | Active                | CALCULATE([Customers], IsActive[Activity]='Active')    | Count of active customers (5,151)                |
| 5   | Has Credit            | CALCULATE([Customers], HasCredit[CreditCard]='Yes')    | Count of customers with credit card (~7,000)     |
| 6   | Inactive NonZero Acct | CALCULATE([Customers], Balance>0, Activity='Inactive') | Inactive customers with money in account (3,105) |
| 7   | Active Zero Balance   | CALCULATE([Customers], Balance=0, Activity='Active')   | Active customers with zero balance (1,873)       |
| 8   | Total Balance         | SUM(Churn_Modelling[Balance])                          | Total money in bank (£765M)                      |
| 9   | Total Salaries        | SUM(Churn_Modelling[EstimatedSalary])                  | Total salary of all customers                    |
| 10  | Average Balance       | AVERAGE(Churn_Modelling[Balance])                      | Mean account balance                             |
| 11  | Average Age           | AVERAGE('Age & Tenure'[Age])                           | Mean customer age                                |
| 12  | Average Salary        | AVERAGE(Churn_Modelling[EstimatedSalary])              | Mean salary (~£100K)                             |
| 13  | Average Tenure        | AVERAGE('Age & Tenure'[Customer_Tenure])               | Mean years with bank                             |
| 14  | Avg. CreditScore      | AVERAGE(Churn_Modelling[CreditScore])                  | Mean credit score (651)                          |
| 15  | Churn Rate            | [Exited]/[Customers]                                   | Percentage who left (20.37%)                     |
| 16  | %Active               | [Active]/[Customers]                                   | Percentage active                                |

| S/<br>N | Measure Name | DAX Formula              | What it does                       |
|---------|--------------|--------------------------|------------------------------------|
| 17      | %HasCredit   | [Has Credit]/[Customers] | Percentage with credit card (~70%) |



## 7.5 Regression in DAX – Line-by-Line Explanation

■ **MOST EXPECTED QUESTION: How was linear regression implemented in DAX? Explain the formula.**

**What is Linear Regression?** It finds the best straight line through data points. Formula:  $y = \beta_1(x) + \beta_0$  where  $\beta_1$  is the slope and  $\beta_0$  is the intercept.

**Slope ( $\beta_1$ ) formula:**  $\beta_1 = \frac{\sum[(x_i - \bar{x})(y_i - \bar{y})]}{\sum[(x_i - \bar{x})^2]}$

**Intercept ( $\beta_0$ ) formula:**  $\beta_0 = \bar{y} - \beta_1 \times \bar{x}$

### Example: Age vs Balance Regression

```
-- Step 1: Create calculated column for x values (Age) age_x = Churn_Modelling[Age]
-- (reference to age from fact table via relationship) -- Step 2: Mean of x mean_age_x =
AVERAGE(Churn_Modelling[age_x]) -- Step 3: Difference from mean for x diff_age_x =
Churn_Modelling[age_x] - Churn_Modelling[mean_age_x] -- Step 4: Balance as y values
bal_y = Churn_Modelling[Balance] -- Step 5: Difference from mean for y diff_bal_y =
Churn_Modelling[bal_y] - AVERAGE(Churn_Modelling[bal_y]) -- Step 6: Product of
differences (numerator component) age_product = Churn_Modelling[diff_age_x] *
Churn_Modelling[diff_bal_y] -- Step 7: Squared differences (denominator component)
diff_age_x_sqrd = Churn_Modelling[diff_age_x] * Churn_Modelling[diff_age_x] -- Step 8:
Numerator measure age_bal_number = SUM(Churn_Modelling[age_product]) -- Step 9:
Denominator measure (age_denominator) = SUM(Churn_Modelling[diff_age_x_sqrd]) -- Step
10: Slope (beta1) -- THE GRADIENT age_bal_slope = [age_bal_number] / [age_denominator] --
RESULT: £565.74 for Exited, £42.11 for Retained -- Step 11: Intercept (beta0) -- WHERE
LINE CROSSES Y-AXIS age_bal_intercept = AVERAGE(Churn_Modelling[bal_y]) -
[age_bal_slope] * AVERAGE(Churn_Modelling[age_x]) -- RESULT: £66K for Exited, £71K for
Retained -- FINAL EQUATION: Balance = £168.42(Age) + £69.93K (overall)
```

### All 6 Regression Results Summary:

| Regression             | Variable X | Variable Y  | Overall Slope | Exited Slope | Retained Slope | Key Finding                        |
|------------------------|------------|-------------|---------------|--------------|----------------|------------------------------------|
| Salary vs Balance      | Salary     | Balance     | +£0.01        | -0.007       | +0.019         | NO significant correlation         |
| Age vs Balance         | Age        | Balance     | +£168.42      | +£565.74     | +£42.11        | SIGNIFICANT — older = more balance |
| Tenure vs Balance      | Tenure     | Balance     | -£264.37      | +£446.30     | -£453.30       | Exited had more balance over time  |
| Balance vs CreditScore | Balance    | CreditScore | 0.00          | £0           | £0             | NO correlation                     |
| Tenure vs CreditScore  | Tenure     | CreditScore | +0.03         | +0.99        | -0.23          | NO significant correlation         |
| Salary vs CreditScore  | Salary     | CreditScore | 0.00          | -£0.0001     | £0             | NO correlation                     |

■ **REMEMBER: Only Age-Balance had SIGNIFICANT correlation (£168.42 per year increase in age). All Credit Score regressions showed NO correlation.**

## PART 8: POWER QUERY & M-LANGUAGE

### 8.1 What is Power Query?

■ **Analogy:** Power Query is like the kitchen where you prepare ingredients before cooking. You wash vegetables (remove nulls), chop them (split columns), and arrange them on a plate (organize the table) before serving (loading to model).

**Power Query** is the data transformation engine in Power BI. It allows you to connect, clean, transform, and shape data before it enters your model. All steps are recorded and repeatable.

### 8.2 What is M-Language?

**M-Language** (also called Power Query Formula Language) is the programming language behind Power Query. Each transformation step you do in Power Query generates M code automatically.

### 8.3 M-Language Examples from the Project:

```
// Creating Age_Group table from scratch using M-Language
let Source =
Table.FromRecords({ [Age_Group="Teenagers", Age_Bracket="18-19 Years", Lower_Age=18,
Upper_Age=19], [Age_Group="Twenties", Age_Bracket="20-29 Years", Lower_Age=20,
Upper_Age=29], [Age_Group="Thirties", Age_Bracket="30-39 Years", Lower_Age=30,
Upper_Age=39], [Age_Group="Forties", Age_Bracket="40-49 Years", Lower_Age=40,
Upper_Age=49], [Age_Group="Fifties", Age_Bracket="50-59 Years", Lower_Age=50,
Upper_Age=59], [Age_Group="Sixties & Above", Age_Bracket="From 60 Years", Lower_Age=60,
Upper_Age=100] }), #"Changed Type" = Table.TransformColumnTypes(Source, {"Age_Group",
type text}, ...), #"Added Index" = Table.AddIndexColumn(#"Changed Type", "Index", 1001,
1, Int64.Type), #"Reordered Cols" = Table.ReorderColumns(...), #"Renamed Columns" =
Table.RenameColumns(#"Reordered Cols", {"Index", "Age_GroupID"}) in #"Renamed Columns"
```

### 8.4 Key Power Query Steps Used in Project:

1. Get Data → Text/CSV → Load original Churn\_Modelling.csv
2. Check column quality (green = valid, yellow = error, red = empty)
3. Check for missing values (all green lines = no nulls)
4. Check for duplicates (all 10,000 rows were unique)
5. Verify data types (text, number, date) for each column
6. Remove unnecessary columns (RowNumber, NumOfProducts removed)
7. Duplicate query → Create dimensional tables (Geography, Gender, IsActive, HasCredit, Churn, Age & Tenure)
8. For Geography: Remove duplicates → Add Index column starting at 11 → Rename as GeographyID
9. For Gender: Same but index starts at 0 (0=Female, 1=Male)
10. For IsActive/HasCredit/Churn: Use Conditional Column to create text categories from 1/0 values
11. Merge Queries: Left Outer Join to combine dimension keys into fact table
12. Remove original categorical columns after merging (avoid redundancy)
13. Create Age\_Group, Tenure\_Class, Salary Range tables via M-Language

■ **EXAM TIP:** The data went from 14 columns to 9 columns in the fact table after pre-processing: 1 primary key (CustomerID) + 5 foreign keys (GeographyID, GenderID, IsActiveID, HasCreditID, ChurnID) + 3 numerical columns (Balance, CreditScore, EstimatedSalary).

## PART 9: XYZ BANK PROJECT – COMPLETE DEEP DIVE

### 9.1 Project Overview

■ **MOST EXPECTED:** Most of your exam questions (5-6 out of 8-10) will come from this project. Know it inside out!

**Project Title:** The Impact of Customer Churn in XYZ Bank

**Business Problem:** XYZ Bank is losing customers (customer churn/attrition). Management needs to understand who is leaving, why they are leaving, and what impact it has on the bank's financial health.

### 9.2 Dataset Information

| Property                 | Value                                                                          |
|--------------------------|--------------------------------------------------------------------------------|
| File Name                | Churn_Modelling.csv                                                            |
| Source                   | Kaggle (by @kmalit — Student Research Assistant, Germany)                      |
| Total Customers          | 10,000 unique customers                                                        |
| Total Columns (Original) | 14 columns                                                                     |
| Numerical Columns        | 6 (Balance, CreditScore, Age, Tenure, EstimatedSalary, RowNumber)              |
| Categorical Columns      | 3 (Geography, Gender, Surname)                                                 |
| Binomial Columns         | 3 (HasCrCard, IsActiveMember, Exited) — values: 0 or 1                         |
| Key Columns              | 2 (RowNumber, CustomerId)                                                      |
| File Size                | 684.86 KB                                                                      |
| Why selected?            | Met business questions + perfect for comparative analysis via binomial columns |

### 9.3 The 5 Business Intelligence Questions (BI Questions)

| Q# | BI Question                                               | Sub-areas                                                                                                                                                                                                   |
|----|-----------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q1 | What is the general knowledge of the customer base?       | a. Distribution: gender, age group, geography b. Engagements: credit card, active, churn, tenure c. Financial: balance, salary, credit score                                                                |
| Q2 | Is there significance between those who left vs remained? | a. Churn across age, gender, geography b. Churn across credit card, activity, accounts, tenure c. Churn vs financial: balance, salary, credit score d. Financial engagements across age group, tenure class |
| Q3 | What contributes to balance of exited vs retained?        | a. Does balance depend on salary? b. How does age affect balance? c. Is there correlation between tenure and balance?                                                                                       |
| Q4 | What influences credit scores of exited vs retained?      | a. Does balance affect credit score? b. Correlation between tenure and credit scores? c. Does salary determine credit score?                                                                                |

| Q# | BI Question                                | Sub-areas                                                                                         |
|----|--------------------------------------------|---------------------------------------------------------------------------------------------------|
| Q5 | What is the impact of customer churn rate? | a. Does it affect customer base? b. Does it affect current balance? c. Does it reduce activeness? |

## 9.4 The 10 Tables Created – Full Description

From 1 flat CSV → 10 structured tables using Power Query + M-Language + DAX. This is SNOWFLAKE SCHEMA.

| Table Name      | Type                             | Key Columns                 | Description                                                         |
|-----------------|----------------------------------|-----------------------------|---------------------------------------------------------------------|
| Churn_Modelling | FACT TABLE                       | CustomerID (PK), 5 FK IDs   | Main table: Balance, CreditScore, EstimatedSalary + foreign keys    |
| Geography       | Dimension (1:*)                  | GeographyID (PK)            | Country: France(11), Spain(12), Germany(13)                         |
| Gender          | Dimension (1:*)                  | GenderID (PK)               | 0=Female, 1=Male                                                    |
| IsActive        | Dimension (1:*)                  | IsActiveID (PK)             | 1=Active, 0=Inactive                                                |
| HasCredit       | Dimension (1:*)                  | HasCreditID (PK)            | 1=Yes (has card), 0=No                                              |
| Churn           | Dimension (1:*)                  | ChurnID (PK)                | 1=Exited, 0=Retained                                                |
| Age & Tenure    | Bridge/Dim (1:1 with Fact)       | CustomerID, Age_GroupID     | Age and Customer_Tenure per customer; bridge to Age_Group           |
| Age_Group       | Dimension (1:* with Age&Tenure;) | Age_GroupID (PK, 1001-1006) | Teenagers/Twenties/Thirties/Forties/Fifties/Sixties&Above;          |
| Salary Range    | Reference Table                  | Salary_Range                | Low (0-33.33%Max), Mid (33-67%), High (67-100%)                     |
| Tenure_Class    | Reference Table                  | Tenure_Group                | New(0-1yr), Fresh(2-3yr), Lasted(4-6yr), Old(7-9yr), Veteran(10+yr) |

■ **SNOWFLAKE pattern in this project: Age\_Group is NOT connected directly to Fact Table. It connects through Age & Tenure table. This chain = Age\_Group → Age & Tenure → Churn\_Modelling = Snowflake!**

## 9.5 Key Findings Summary (Exam-Ready)

■ **These numbers WILL come in your exam. Memorize them!**

| Finding                       | Number/Value    | Insight                            |
|-------------------------------|-----------------|------------------------------------|
| Total customers               | 10,000          | All unique CustomerIDs             |
| Exited customers              | 2,037 (20.37%)  | 1 in 5 customers left the bank     |
| Retained customers            | 7,963 (79.63%)  | 4 in 5 stayed                      |
| Customers with credit card    | ~7,000 (70%)    | 7 in 10 have credit card           |
| Customers without credit card | ~3,000 (30%)    | 3 in 10 have no card               |
| Active customers              | 5,151 (51.51%)  | Slightly more than half are active |
| Customers >3 years tenure     | ~60%            | Majority are veteran customers     |
| Largest age group             | Thirties (4.3K) | Almost half of customer base       |
| Smallest age group            | Teenagers       | Least represented                  |

| Finding                              | Number/Value            | Insight                                              |
|--------------------------------------|-------------------------|------------------------------------------------------|
| Geography with most customers        | France (~50%)           | Half the base is in France                           |
| Germany churn vs France churn        | 814 vs 810              | Germany churns more despite smaller base             |
| Female churn > Male churn            | ~1,100 vs ~900          | Females leave more                                   |
| Age group with most churn            | Forties (806)           | Middle-aged customers leave most                     |
| Total Balance                        | £765 Million            | Overall bank portfolio                               |
| Average Salary                       | ~£100,000               | Similar for both Exited and Retained                 |
| Average Credit Score                 | 651 (both groups)       | No difference in credit scores                       |
| Age-Balance correlation              | £168.42 per year of age | SIGNIFICANT — older = richer                         |
| Exited age-balance slope             | £565.74 per year        | Exited customers' balance grows much faster with age |
| Tenure-Balance correlation (overall) | -£264.37 per year       | NEGATIVE — longer tenure = lower balance             |
| Exited tenure-balance slope          | +£446.30                | Exited kept MORE money over time                     |
| Retained tenure-balance slope        | -£453.30                | Retained customers' balance DECREASES over time      |

## 9.6 Recommendations from the Project

These recommendations are exam gold — show examiner you understand business implications:

1. Since churn rate does NOT affect activeness, focus on converting inactive accounts to active SEPARATELY — not just by reducing churn.
2. Launch marketing campaigns against ZERO account balance — data shows tendency to leave INCREASES when balance hits zero.
3. Target OLDER customers for acquisition — positive correlation between age and balance means older customers bring more money.
4. It is okay to launch a zero-balance account opener for new customers — it grows the base. But have strategies to grow those accounts.
5. Revisit or promote the VALUE of credit cards — majority of those who LEFT had credit cards (more than twice those without).

## PART 10: DASHBOARD WALKTHROUGH & CHART ANALYSIS

### 10.1 Dashboard 1: Home Page

The Home page was a navigation hub with infographics showing key insights. It included page buttons for seamless navigation, button icons as visual clues, and a colour theme of Gold, Wine Red, and Grey.

### 10.2 Dashboard 2: Customer Distribution

Answers BI Question 1. KPIs shown at top: 10K Customers, 2,037 Exited, £765M Total Balance, 651 Avg Credit Score, £100K Avg Salary.

#### Charts Used in Customer Distribution Dashboard:

| Visual/Chart              | Type                 | Why This Chart?                                                         | What It Shows                                   | Potential Exam Question                                  |
|---------------------------|----------------------|-------------------------------------------------------------------------|-------------------------------------------------|----------------------------------------------------------|
| Customers by Gender       | Pie Chart            | Shows proportional distribution of categories (Male vs Female)          | 54.5% Male, 45.5% Female — ~10% gap             | Why use a pie chart for gender distribution?             |
| Customers by Age_Group    | Horizontal Bar Chart | Compares quantities across categories using bar lengths                 | Thirties = 4.3K (largest); Teenagers = smallest | Why horizontal bar for age group, not pie?               |
| Customers by Geography    | Map Visual           | Geographic data best shown on actual map with bubble size = quantity    | France=50%, Germany=25%, Spain=25%              | What does bubble size represent on the map?              |
| Churn Rate                | KPI Donut Chart      | Shows part-to-whole ratio (20% exited, 80% retained)                    | 20.37% churn rate                               | Why use a donut chart for churn rate?                    |
| Active Customers          | Gauge (Axis) Chart   | Shows progress against a scale/target — proportion of active vs total   | 5,151 active out of 10K (51.5%)                 | When would you use a gauge chart vs a donut?             |
| Customers by CreditCard   | Donut Chart          | Part-to-whole: Yes/No credit card                                       | 70% Yes, 30% No                                 | What is the difference between pie and donut?            |
| Customers by Tenure Class | Aster Plot Chart     | Shows distribution across multiple categorical groups using blade sizes | ~60% customers >3 years tenure                  | What type of data is best visualized with an Aster Plot? |
| Customers by Account Type | 100% Stacked Bar     | Shows proportional breakdown within categories                          | Active Zero, Inactive NonZero, Active NonZero   | Why 100% stacked bar for account type?                   |
| Salary by Salary_Group    | Table                | Detailed breakdown: Low/Mid/High salary with counts                     | Nearly equal across Low/Mid/High (3,303-3,383)  | Why use a table for salary group rather than a chart?    |



## 10.3 Dashboard 3: Churn Distribution (BI Question 2)

| Visual                                   | Type                   | Why This Type?                                                                                                         | Key Finding                                                   |
|------------------------------------------|------------------------|------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|
| Age_Group vs ChurnStatus                 | Horizontal Stacked Bar | Compare churn across categories — stacked shows both Exited & Retained in one bar; horizontal best for many categories | Forties age group has most churn (806 exited)                 |
| Gender vs ChurnStatus                    | Vertical Stacked Bar   | Compare across fewer categories (M/F) — vertical works for 2 categories                                                | Females leave more (1,100 vs 900)                             |
| Country vs ChurnStatus                   | Horizontal Stacked Bar | 3 countries — stacked shows which country churns more proportionally                                                   | Germany: 814 leavers; France: 810 despite larger base         |
| Credit by ChurnStatus                    | Donut Chart            | Part-to-whole: shows what % of credit card holders left                                                                | Out of 7K with credit card, 1K left                           |
| Active by ChurnStatus                    | Donut Chart            | Part-to-whole: shows what % of active customers left                                                                   | Out of 5K active, 1K left (14.2%)                             |
| Active Zero vs Inactive NonZero by Churn | Aster Plot             | Two dual measures across churn status — blade size shows magnitude                                                     | 1,873 active zero: 180 left. 3,105 inactive nonzero: 982 left |
| Tenure by ChurnStatus                    | Violin Plot            | Statistical distribution — shows spread, density, and outliers better than box plots for continuous numerical data     | Exited: 2-8 years. Retained: 3-7 years. Both average ~5 years |
| Average Balance by ChurnStatus           | Box and Whisker Plot   | Shows statistical summary: min, Q1, median, Q3, max — better than simple bar for financial distributions               | Exited had £20K more average balance                          |
| Average Salary by ChurnStatus            | Box and Whisker Plot   | Same reasoning — shows spread of salaries, not just average                                                            | No difference — both near £100K                               |
| Score by ChurnStatus                     | Violin Plot            | Distribution comparison — credit score distributions nearly identical                                                  | Both at ~645-651 average — no difference                      |

## 10.4 Dashboard 4: Churn Bivariate Analysis (BI Question 2d)

**Bivariate analysis** means looking at TWO numerical variables together (plus a category = ChurnStatus as colour/legend). **Scatter plots** are always used for bivariate analysis.

| Chart                                 | X axis          | Y axis          | Category               | Key Finding                                                  |
|---------------------------------------|-----------------|-----------------|------------------------|--------------------------------------------------------------|
| CreditScore vs Age                    | Age             | CreditScore     | ChurnStatus            | No clear disparity — older exited had slightly lower scores  |
| CreditScore vs Tenure                 | Customer_Tenure | CreditScore     | ChurnStatus            | Near-equal, difference of ~10 points — no significance       |
| Balance vs Salary by Tenure           | Avg Salary      | Avg Balance     | Customer_Tenure labels | Exited clearly have more balance at every tenure point       |
| Balance vs Age by Age_Group           | Avg Age         | Avg Balance     | Age_Group labels       | Exited had higher balance across all age groups              |
| Balance vs Tenure by Tenure_Class     | Avg Tenure      | Avg Balance     | Tenure_Class labels    | Exited balance always above £80K; Retained always below £80K |
| CreditScore vs Tenure by Tenure_Class | Avg Tenure      | Avg CreditScore | Tenure_Class           | Retained had slightly higher scores, but not by much         |

## 10.5 Dashboard 5 & 6: Balance Regression & Credit Score Regression

These dashboards show 3 charts each: (1) Scatter plot with trend line equation, (2) Slope by ChurnStatus bar chart, (3) Intercept by ChurnStatus bar chart.

Each regression dashboard shows the mathematical relationship between two variables, split by Exited vs Retained customers.

## 10.6 Dashboard 7: AI Insights (BI Question 5)

| AI Chart                                     | What it does                                                                  | Key Finding                                                                          |
|----------------------------------------------|-------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| Key Influencer Chart (Exited to Increase)    | AI automatically identifies what factors cause Exited to increase             | Balance = 0 or less increases Exited count by 0.74                                   |
| Key Influencer Chart (Customers to Increase) | AI identifies what makes total customers grow                                 | Balance = 0 or less increases customers by 0.02; CreditScore > 528 increases by 0.01 |
| Key Influencer Chart (Retained to Decrease)  | What causes retained to decrease                                              | Balance > 0 decreases retained by 0.64; CreditCard=No; CreditScore < 480             |
| Key Influencer Chart (Active to Increase)    | What makes customers more active                                              | CreditScore > 849 increases Active by 18.69; Geography=France increases by 1.29      |
| Decomposition Tree                           | Breaks down 10,000 customers by: ChurnStatus → CreditCard → Activity → Gender | Exited=2,037 → Yes<br>CreditCard=1,424 →<br>Inactive=942 → Female=523                |

■ **KEY CONCLUSION from AI Insights: Churn status has the MOST IMPORTANCE in affecting the overall customer base among all dimensions (ChurnStatus, CreditCard, Activity, Gender).**

## PART 11: REGRESSION & NUMERIC PROBLEMS – STEP BY STEP

### 11.1 How to Calculate Linear Regression Slope — Step by Step

**Formula:**  $\beta_1 = \frac{\sum[(x_i - \bar{x})(y_i - \bar{y})]}{\sum[(x_i - \bar{x})^2]}$  and  $\beta_0 = \bar{y} - \beta_1 \times \bar{x}$

**Worked Example:** Suppose we have 5 customers with ages and balances:

| Customer | Age (x)      | Balance (y)       | $x - \bar{x}$ | $y - \bar{y}$  | $(x - \bar{x})(y - \bar{y})$ | $(x - \bar{x})^2$ |
|----------|--------------|-------------------|---------------|----------------|------------------------------|-------------------|
| 1        | 25           | £60,000           | 25-40 = -15   | 60K-76K = -16K | (-15)×(-16K) = +240K         | 225               |
| 2        | 35           | £70,000           | 35-40 = -5    | 70K-76K = -6K  | (-5)×(-6K) = +30K            | 25                |
| 3        | 40           | £75,000           | 40-40 = 0     | 75K-76K = -1K  | (0)×(-1K) = 0                | 0                 |
| 4        | 45           | £80,000           | 45-40 = +5    | 80K-76K = +4K  | (5)×(4K) = +20K              | 25                |
| 5        | 55           | £95,000           | 55-40 = +15   | 95K-76K = +19K | (15)×(19K) = +285K           | 225               |
| TOTAL    | $\bar{x}=40$ | $\bar{y}=£76,000$ | —             | —              | $\Sigma = 575K$              | $\Sigma = 500$    |

**Step 1:** Calculate mean of x:  $\bar{x} = (25+35+40+45+55)/5 = 200/5 = 40$

**Step 2:** Calculate mean of y:  $\bar{y} = (60K+70K+75K+80K+95K)/5 = 380K/5 = £76,000$

**Step 3:** Calculate  $(x - \bar{x})$  and  $(y - \bar{y})$  for each row (see table above)

**Step 4:** Multiply them  $\rightarrow \sum(x_i - \bar{x})(y_i - \bar{y}) = 240K + 30K + 0 + 20K + 285K = 575,000$

**Step 5:** Square each  $(x - \bar{x}) \rightarrow \sum(x_i - \bar{x})^2 = 225 + 25 + 0 + 25 + 225 = 500$

**Step 6:** Slope  $\beta_1 = 575,000 / 500 = 1,150$  (£1,150 increase in balance per year of age)

**Step 7:** Intercept  $\beta_0 = 76,000 - 1,150 \times 40 = 76,000 - 46,000 = £30,000$

**Final Equation:** Balance = £1,150(Age) + £30,000

**COMMON MISTAKES in Regression:** (1) Forgetting to calculate the mean first. (2) Not squaring the denominator  $(x_i - \bar{x})^2$ . (3) Confusing slope with intercept. (4) Using population formula instead of sample formula.

### 11.2 Interpreting Correlation Results

| Slope Value                     | Interpretation                                     |
|---------------------------------|----------------------------------------------------|
| Positive slope (+)              | As X increases, Y increases (positive correlation) |
| Negative slope (-)              | As X increases, Y DECREASES (negative correlation) |
| Slope near 0 ( $\approx 0.00$ ) | NO meaningful correlation — X does not predict Y   |
| Large slope magnitude           | Strong relationship — X has big impact on Y        |
| Small slope magnitude           | Weak relationship — X has little impact on Y       |

**From the project:** Age-Balance slope = +£168.42 (SIGNIFICANT positive). Salary-Balance slope = +£0.01 (NEGLIGIBLE — no real correlation). Tenure-Balance slope = -£264.37 (SIGNIFICANT NEGATIVE).

## PART 12: SQL TABULAR MODEL & AGILE BI (RESEARCH PAPER)

### 12.1 Paper Overview

**Paper Title:** SQL Server Tabular Model: A Step Towards Agile BI

**Author:** Stevan Mrdalj, Eastern Michigan University (2013)

**Main Argument:** Traditional BI is too slow. The new SQL Server Tabular Model uses in-memory columnar storage (xVelocity) to enable fast, flexible, self-service BI — a step towards Agile BI.

### 12.2 The xVelocity Engine — How Columnar Storage Works

■ **Analogy:** Normal databases store data ROW by row (like reading a book line by line). xVelocity stores data COLUMN by column (like reading every 3rd word in the entire book at once). For analytics that need 'sum of all balances', column storage is MUCH faster.

**Steps:** (1) Convert rows of every table into columns. (2) Encode and compress each column independently. (3) Store all values from one column in a single memory block. (4) Query runs as brute-force scan in main memory — no indexes needed.

### 12.3 Two Types of xVelocity Compression

| Compression Type          | How it works                                                                                             | Example                                                                             |
|---------------------------|----------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Run Length Encoding (RLE) | Replaces repeated consecutive values with value + count of repetitions                                   | MI, MI, MI, MI → MI [start=1, count=4]                                              |
| Dictionary Encoding (DE)  | Creates a dictionary of all distinct values with IDs. Stores ID numbers instead of repeated text strings | Detroit=0, Lansing=1, Toledo=2.<br>Then stores: 0,0,1,1,2,2,2 instead of city names |

### 12.4 Three Criteria for Agile BI (from the paper)

1. **Ability to integrate new and unforeseen data sources** — The Tabular Model can source from relational DBs, OLAP cubes, flat files, Excel, OData feeds, cloud services. Traditional OLAP only allows one relational DB.
2. **Ability to rapidly embrace change** — No pre-computed aggregations, new rows can be added in real-time, queries dynamically calculated on the fly.
3. **Ability to provide self-service BI** — Business analysts can start with PowerPivot for Excel (personal BI) and scale to SQL Server Data Tools (corporate BI) using the SAME model.

### 12.5 Limitations of the Tabular Model

- Only supports regular and role-playing relationships (1 role-playing active at a time)
- Does NOT support many-to-many relationships natively
- Does NOT support parent-child hierarchy natively
- No built-in time dimension
- No semi-additive functions (powerful aggregations)
- Main memory is VOLATILE — data recovery strategies critical
- Not designed for cloud computing at time of paper

### 12.6 BI Semantic Model (BISM) Architecture

| Layer          | Multidimensional                   | Tabular                                        |
|----------------|------------------------------------|------------------------------------------------|
| Data Model     | Multidimensional model             | Tabular model (tables, columns, relationships) |
| Business Logic | MDX (MultiDimensional eXpressions) | DAX (Data Analysis Expressions)                |
| Data Access    | ROLAP / MOLAP                      | xVelocity (in-memory) / Direct Query           |

■ **EXAM TIP: Key difference between Tabular and Multidimensional: Tabular uses DAX + xVelocity. Multidimensional uses MDX + ROLAP/MOLAP. Both live in BISM.**

## PART 13: MOST EXPECTED EXAM QUESTIONS + MODEL ANSWERS

### Section A: Short Questions (2-5 marks each)

#### ■ Q1: What is Business Intelligence?

**Model Answer:** BI is a set of strategies and technologies used to analyze raw business data and present actionable information to help decision-makers. Key terms: data-driven decisions, analytics, dashboards, KPIs.

#### ■ Q2: Define Customer Churn.

**Model Answer:** Customer churn (attrition) is the rate at which customers stop doing business with an entity.  $\text{Churn Rate} = (\text{Customers who left} / \text{Total Customers}) \times 100$ . In the project, churn rate = 20.37%.

#### ■ Q3: What is a KPI?

**Model Answer:** KPI (Key Performance Indicator) is a measurable value that shows how effectively an organization achieves its key objectives. Example: Churn Rate, Total Balance, Average Credit Score.

#### ■ Q4: What is a Dashboard?

**Model Answer:** A dashboard is a visual display of the most important metrics needed to achieve objectives, consolidated on a single screen for at-a-glance monitoring. Example: Power BI dashboard with customer KPIs.

#### ■ Q5: What is a Fact Table?

**Model Answer:** A fact table is the central table in a data warehouse that contains quantitative data (measures/metrics) and foreign keys linking to dimension tables. Example: Churn\_Modelling with Balance, CreditScore, Salary.

#### ■ Q6: What is a Dimension Table?

**Model Answer:** A dimension table contains descriptive/qualitative attributes used to filter, group, or label data in analyses. Example: Geography (France/Spain/Germany), Gender (Male/Female).

#### ■ Q7: What is the difference between a Measure and a Calculated Column?

**Model Answer:** Measure: Calculated at query time, not stored in table, uses filter context, no memory impact. Calculated Column: Calculated at refresh time, stored permanently in table, uses row context, adds to model size.

#### ■ Q8: What is a View in SQL/Power BI?

**Model Answer:** A view is a stored SQL query that doesn't physically store data. Every time accessed, it re-executes the query showing current data. Benefits: simplifies joins, enables security, ensures naming consistency.

#### ■ Q9: What is xVelocity?

**Model Answer:** xVelocity is Microsoft's in-memory columnar storage engine used in the Tabular Model. It stores data column-by-column with Run Length Encoding and Dictionary Encoding compression for ultra-fast analytics without pre-built indexes.

### ■ Q10: What is OLAP?

**Model Answer:** OLAP (Online Analytical Processing) enables fast, multidimensional analysis of business data. It allows users to 'slice and dice' data across dimensions like time, product, and geography. Types: MOLAP, ROLAP, HOLAP.

## Section B: Long/Comparison Questions (10-15 marks each)

### ■ L1: Compare Power BI and Tableau in detail.

**Key Points to Cover:** Intro: Both are BI tools for data visualization. Key points — Cost (PBI free/Pro, Tableau expensive). Ease of Use (PBI easier for Microsoft users, Tableau steeper learning). Performance (Tableau better with huge datasets). AI (both have AI, PBI uses Azure ML). Data Prep (PBI Power Query integrated, Tableau uses separate Tableau Prep). Scripting (PBI uses DAX+M, Tableau uses Python/R). Best for (PBI: Microsoft ecosystem, Tableau: complex custom visuals). Conclude: Choice depends on budget and needs.

### ■ L2: Explain Import, Direct Query, and Live Connection in Power BI with their use cases.

**Key Points to Cover:** Import: Copies data into PBI memory. Fastest. Good for small-medium datasets. Full DAX. Needs scheduled refresh. Direct Query: No data import, queries DB in real-time every time. Good for large/real-time data. Limited DAX. Slower performance. Live Connection: Connects to existing SSAS/Analysis Services model. No transformation. For enterprise shared models. Give use cases: Import for most projects. Direct Query for financial dashboards needing real-time. Live Connection for large organizations with existing SSAS investment. Exam keywords: performance, data freshness, scalability, transformation.

### ■ L3: Explain the data modeling approach used in the XYZ Bank project.

**Key Points to Cover:** Start: Original flat CSV with 14 columns, 10,000 rows. Approach: Snowflake Schema (not Star, because Age\_Group connects through Age & Tenure bridge table). Created 10 tables: 1 Fact (Churn\_Modelling) + 6 Dimensional (Geography, Gender, IsActive, HasCredit, Churn, Age & Tenure, Age\_Group) + 2 Reference tables (Salary\_Range, Tenure\_Class). Relationships: Mostly 1:\* (one-to-many). Age & Tenure has 1:1 with Fact. Age\_Group connects to Age & Tenure (bridge). Keys: All dimensional tables have primary keys (GeographyID, GenderID, etc.) connected to foreign keys in Fact table. Benefits: Better performance, less redundancy, correct relationships for DAX aggregations.

### ■ L4: What are the key findings of the XYZ Bank customer churn analysis?

**Key Points to Cover:** Structure findings by BI question: Q1 (base): 10K customers, 20.37% churn (2,037 exited), 70% have credit card, 60% tenure >3 years, France has most customers. Q2 (comparison): Forties age group churns most (806), Females churn more, Germany proportionally higher churn than France, Exited had £20K more average balance. Q3 (balance factors): Age-Balance significant (+£168.42/year), Salary-Balance insignificant, Tenure-Balance negative overall (-£264.37) but positive for Exited (+£446.30). Q4 (credit score): Nothing significantly correlates with credit score — not balance, tenure, or salary. Q5 (churn impact): Affects customer base (churn status has most importance per decomposition tree), does affect balance (exited had more), does NOT affect activeness. Recommendations: Target older customers, combat zero balances, review credit card value.

## Section C: Project-Based Questions

**P1.** Why was a Snowflake Schema used instead of Star Schema in the project? — Because Age\_Group is not directly connected to the Fact Table; it connects through the Age & Tenure bridge table, creating a multi-level chain = Snowflake.

- P2.** What DAX function was used to count unique customers, and why not COUNT? — DISTINCTCOUNT was used to ensure each CustomerID is counted only ONCE, even if it appeared multiple times. COUNT would count all rows including potential duplicates.
- P3.** Explain why Violin Plot was used for Tenure and Credit Score comparisons. — Violin plots show the FULL statistical distribution (density, spread, quartiles) of a numerical variable across categories — better than box plots for showing where data clusters.
- P4.** What is the significance of Balance = 0? — The AI Key Influencer chart found that Balance = 0 is the TOP factor causing customers to EXIT. Also, balance = 0 is a top factor increasing overall customer count (zero-balance accounts attract new signups).
- P5.** Why was the Aster Plot used for Tenure Class? — Aster Plot shows distribution across multiple categorical groups using blade sizes, making it visually effective for comparing across many tenure classes (0-1yr, 2-3yr, etc.).
- P6.** What does the Decomposition Tree show? — It shows how 10,000 customers break down hierarchically by ChurnStatus → CreditCard → Activity → Gender. Key insight: ChurnStatus has most importance in the breakdown.
- P7.** Why was Age & Tenure kept as a SEPARATE table? — Because each CustomerID had UNIQUE age and tenure values — no duplicates. Removing duplicates would cause data loss. It needed its own table with CustomerID as key.
- P8.** What were the 3 numerical columns kept in the Fact Table after pre-processing? — Balance, CreditScore, EstimatedSalary. (Age and Tenure were moved to the Age & Tenure table.)



## Section D: Top 30 MCQ Practice

| #  | Question                                                  | A                         | B                                             | C                        | D                              |
|----|-----------------------------------------------------------|---------------------------|-----------------------------------------------|--------------------------|--------------------------------|
| 1  | Which connection mode in Power BI stores data in-memory?  | A) Direct Query           | B) Live Connection                            | C) Import ✓              | D) Native Query                |
| 2  | What is the Churn Rate in the XYZ Bank project?           | A) 10.37%                 | B) 20.37% ✓                                   | C) 30.37%                | D) 50%                         |
| 3  | Which schema was used in the XYZ Bank project?            | A) Star Schema            | B) Snowflake Schema ✓                         | C) Galaxy Schema         | D) Flat Schema                 |
| 4  | DISTINCTCOUNT is used to:                                 | A) Sum values             | B) Count all rows                             | C) Count unique values ✓ | D) Calculate average           |
| 5  | Which DAX function evaluates an expression with a filter? | A) SUM                    | B) CALCULATE ✓                                | C) VALUES                | D) AVERAGE                     |
| 6  | A Fact Table contains:                                    | A) Descriptive data       | B) Numerical measures & foreign keys ✓        | C) Only primary keys     | D) Reference data              |
| 7  | How many tables were in the final data model?             | A) 5                      | B) 8                                          | C) 10 ✓                  | D) 14                          |
| 8  | What is xVelocity?                                        | A) A programming language | B) In-memory columnar storage engine ✓        | C) A chart type          | D) A data source               |
| 9  | A database view physically stores data.                   | A) True                   | B) False ✓                                    |                          |                                |
| 10 | Which chart best shows part-to-whole distribution?        | A) Bar chart              | B) Line chart                                 | C) Donut/Pie chart ✓     | D) Scatter plot                |
| 11 | Direct Query mode is best for:                            | A) Small static datasets  | B) Real-time large data ✓                     | C) Offline analysis      | D) Excel integration           |
| 12 | The slope in Age vs Balance regression was:               | A) -£168.42               | B) +£0.01                                     | C) +£168.42 ✓            | D) -£264.37                    |
| 13 | Which language is used in Power Query?                    | A) DAX                    | B) MDX                                        | C) M-Language ✓          | D) SQL                         |
| 14 | What is the percentage of male customers in the bank?     | A) 45.5%                  | B) 54.5% ✓                                    | C) 60%                   | D) 70%                         |
| 15 | Snowflake Schema differs from Star Schema because:        | A) It has no fact table   | B) Dimension tables are normalized/branched ✓ | C) It uses fewer tables  | D) It doesn't use foreign keys |
| 16 | Which age group had the most customer churn?              | A) Thirties               | B) Twenties                                   | C) Forties ✓             | D) Fifties                     |

| #  | Question                                                                               | A                          | B                                      | C                                                       | D                         |
|----|----------------------------------------------------------------------------------------|----------------------------|----------------------------------------|---------------------------------------------------------|---------------------------|
| 17 | A Violin Plot is best for:                                                             | A) Geographic distribution | B) Part-to-whole comparison            | C) Numerical distribution across categories ✓           | D) Timeline visualization |
| 18 | Total bank balance in the project was approximately:                                   | A) £76.5M                  | B) £765M ✓                             | C) £7.65B                                               | D) £100K                  |
| 19 | Which factor had the most importance in the customer base decomposition?               | A) Gender                  | B) Activity                            | C) CreditCard                                           | D) ChurnStatus ✓          |
| 20 | M-Language is used in:                                                                 | A) DAX calculations        | B) Power Query transformations ✓       | C) SQL Server                                           | D) Dashboard design       |
| 21 | What was the churn rate for the 'Forties' age group (exited from 2.6K)?                | A) ~15%                    | B) ~31% ✓                              | C) ~50%                                                 | D) ~20%                   |
| 22 | Average salary of both Exited and Retained customers was:                              | A) Very different          | B) Same (~£100K) ✓                     | C) Exited earned more                                   | D) Retained earned more   |
| 23 | RLE stands for:                                                                        | A) Real-time Lookup Engine | B) Run Length Encoding ✓               | C) Relational Logic Engine                              | D) Row Level Encryption   |
| 24 | Agile BI criteria from research paper: How many criteria?                              | A) 2                       | B) 3 ✓                                 | C) 4                                                    | D) 5                      |
| 25 | Which table in the project served as the bridge table?                                 | A) Geography               | B) Churn                               | C) Age & Tenure ✓                                       | D) Salary Range           |
| 26 | What does CALCULATE do in DAX?                                                         | A) Sums values             | B) Filters and evaluates expressions ✓ | C) Creates new tables                                   | D) Counts rows            |
| 27 | A Box and Whisker plot shows:                                                          | A) Percentage breakdown    | B) Geographic distribution             | C) Statistical summary (min, Q1, median, Q3, max) ✓     | D) Time series trends     |
| 28 | Why are Views preferred over Tables in Power BI (main reason)?                         | A) Views are faster        | B) Views store more data               | C) Views ensure naming consistency and simplify joins ✓ | D) Views never change     |
| 29 | Which country had proportionally the highest churn rate despite smaller customer base? | A) France                  | B) Spain                               | C) Germany ✓                                            | D) All equal              |

| #  | Question                                                | A                               | B                                                                            | C                         | D                    |
|----|---------------------------------------------------------|---------------------------------|------------------------------------------------------------------------------|---------------------------|----------------------|
| 30 | Live Connection is different from Direct Query because: | A) Live Connection imports data | B) Live Connection uses existing SSAS models without any PBI data modeling ✓ | C) Direct Query is faster | D) They are the same |

■ Answers marked with ✓ are correct.

## PART 14: FINAL REVISION – ONE-NIGHT CHEAT SHEET ■

■ IF YOU ONLY READ THIS SECTION, YOU CAN STILL PASS ■

### 14.1 Top 20 Definitions to Memorize

**Business Intelligence:** Using data tools & strategies to analyze business data and support decision-making.

**Customer Churn:** Rate at which customers leave a company. Formula:  $\text{Exited} / \text{Total Customers} \times 100$ .

**KPI:** Key Performance Indicator — a measurable value showing achievement of objectives. e.g., Churn Rate.

**Dashboard:** A visual display of key metrics on one screen for quick decision support.

**Power BI:** Microsoft's business analytics service for creating interactive visualizations and reports.

**Tableau:** Data visualization tool with drag-and-drop interface; strong on customization but expensive.

**Import Mode:** Default PBI mode — copies data into memory. Fastest but not live.

**Direct Query:** Real-time queries to source database every time visual updates. No data stored in PBI.

**Live Connection:** Uses existing Analysis Services model. No data import or transformation.

**Fact Table:** Central table with numerical measures (Balance, Sales) and foreign keys linking to dimensions.

**Dimension Table:** Descriptive lookup table (Geography, Gender) used for filtering and grouping.

**Star Schema:** Fact table surrounded by flat dimension tables — simple, fast.

**Snowflake Schema:** Like star but dimensions are normalized — sub-dimensions branch off main dimensions.

**DAX:** Data Analysis Expressions — formula language for measures and calculated columns in Power BI.

**Measure:** DAX calculation evaluated at query time (dynamic). e.g.,  $\text{Churn Rate} = [\text{Exited}] / [\text{Customers}]$ .

**Calculated Column:** DAX column added permanently to a table at refresh time. e.g., Salary\_Group.

**Power Query / M-Language:** ETL tool in Power BI for connecting, cleaning, and transforming data before loading.

**View (SQL):** Stored SQL query that doesn't physically store data — always shows current data from underlying tables.

**xVelocity:** Microsoft's in-memory columnar storage engine using Run Length Encoding and Dictionary Encoding.

**Agile BI:** Fast, flexible BI that allows self-service, integrates any data source, and embraces change quickly.

### 14.2 Critical Numbers to Remember

| Number         | What it means                     |
|----------------|-----------------------------------|
| 10,000         | Total customers in dataset        |
| 2,037 (20.37%) | Customers who exited / Churn Rate |
| 7,963 (79.63%) | Customers retained                |
| 5,151 (51.51%) | Active customers                  |
| ~7,000 (70%)   | Customers with credit card        |

| Number        | What it means                                                     |
|---------------|-------------------------------------------------------------------|
| £765M         | Total balance in bank                                             |
| £100K         | Average salary (both groups, no difference)                       |
| 651           | Average credit score (both groups, no difference)                 |
| 4,300         | Customers in 'Thirties' (largest group)                           |
| 806           | Exited customers in 'Forties' (most churn by age group)           |
| 814 vs 810    | Germany vs France churners (Germany proportionally higher)        |
| £168.42/year  | Age-Balance regression slope (overall — SIGNIFICANT)              |
| £565.74/year  | Age-Balance slope for EXITED customers (much stronger)            |
| -£264.37/year | Tenure-Balance slope (overall — NEGATIVE, longer = lower balance) |
| +£446.30      | Tenure-Balance slope for EXITED (positive — they kept money)      |
| 0.00          | All Credit Score regression slopes (NO correlation with anything) |
| 10            | Number of tables in the final data model                          |
| 14 → 9        | Columns in fact table: original 14 → after cleaning 9             |
| 3 criteria    | Agile BI criteria from research paper                             |

## 14.3 Top 5 Comparison Tables (Quick Reference)

### FACT vs DIMENSION:

| Property     | Fact Table                       | Dimension Table          |
|--------------|----------------------------------|--------------------------|
| Data         | Numbers/measures                 | Descriptive text         |
| Size         | Large                            | Small                    |
| Keys         | Foreign Keys (FK)                | Primary Key (PK)         |
| Example      | Churn_Modelling (Balance, Score) | Geography, Gender, Churn |
| Aggregation? | YES (SUM, AVG, COUNT)            | NO                       |

### MEASURE vs CALCULATED COLUMN:

| Property | Measure                           | Calculated Column           |
|----------|-----------------------------------|-----------------------------|
| Timing   | Query time (dynamic)              | Refresh time (static)       |
| Stored?  | No                                | Yes (in table)              |
| Context  | Filter context                    | Row context                 |
| Memory?  | No impact                         | Adds to model size          |
| Example  | Churn Rate = [Exited]/[Customers] | Salary_Group = Low/Mid/High |

### IMPORT vs DIRECT QUERY vs LIVE (Quickest):

|              | Import       | Direct Query | Live Connection |
|--------------|--------------|--------------|-----------------|
| Data in PBI? | YES          | NO           | NO              |
| Live data?   | NO (refresh) | YES          | YES             |
| Performance  | Fastest      | Slower       | Good            |
| Full DAX?    | YES          | Limited      | Limited         |

### STAR vs SNOWFLAKE:

|            | Star                  | Snowflake               |
|------------|-----------------------|-------------------------|
| Dimensions | Flat (not normalized) | Normalized (sub-tables) |
| Joins      | Fewer                 | More                    |
| Speed      | Faster                | Slower                  |
| Project    | Base structure        | Used (Age_Group branch) |

## 14.4 Top 10 DAX Formulas to Remember

**Customers** = DISTINCTCOUNT(Churn\_Modelling[CustomerId]) → Count unique customers

**Exited** = CALCULATE([Customers], Churn[ChurnStatus]='Exited') → Count who left

**Active** = CALCULATE([Customers], IsActive[Activity]='Active') → Count active

**Has Credit** = CALCULATE([Customers], HasCredit[CreditCard]='Yes') → Count with credit card

**Total Balance** = SUM(Churn\_Modelling[Balance]) → Sum all balances

**Avg CreditScore** = AVERAGE(Churn\_Modelling[CreditScore]) → Mean credit score

**Churn Rate** = [Exited] / [Customers] → Percentage who left

**Inactive NonZero Acct** = CALCULATE([Customers], Balance>0, Activity='Inactive') → Inactive with money

**Slope (generic)** = [Numerator\_Measure] / [Denominator\_Measure] → Linear regression slope

**Intercept (generic)** = AVERAGE([y\_column]) - [slope\_measure] \* AVERAGE([x\_column]) → Regression intercept

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## 14.5 Key Insights You MUST Know for the Exam

- Churn rate = 20.37% — 1 in 5 customers left XYZ Bank.
- Balance = 0 is the #1 factor causing customers to leave (AI Key Influencer finding).
- Churn status is the MOST IMPORTANT factor in the customer base decomposition.
- Churn rate does NOT affect customer activeness.
- Only Age-Balance showed SIGNIFICANT correlation (£168.42 per year).
- Credit Score has NO significant correlation with Balance, Tenure, or Salary.
- Exited customers had MORE balance than Retained — by ALL age groups and tenure classes.
- 10 tables created from 1 flat CSV — Snowflake Schema.
- The project used 17 measures + many calculated columns + 3 M-Language tables.
- Views are preferred over tables for consistency, security, and join simplification.
- Import mode is default and fastest; Direct Query is for real-time large data.
- Tabular Model criteria for Agile BI: integrate sources, embrace change, self-service.
- xVelocity uses Run Length Encoding + Dictionary Encoding for columnar compression.
- BI Semantic Model (BISM) = Multidimensional (MDX) + Tabular (DAX) under one roof.
- Scatter plots are always used for bivariate analysis in the project.

■ **GOOD LUCK IN YOUR EXAM! You have this!** ■

*Prepared from: BI Research Paper (Mrdalj 2013), XYZ Bank ICA Report, Power BI vs Tableau notes, Import/DirectQuery/LiveConnection notes, Native DB Query notes, Views vs Tables notes, Power BI Server notes.*